

4(a)	any four from: 1 marker gene added with gene of interest (GOI) ; 2 use same promoter ; 3 (so) both genes, transcribed / expressed (together) ; 4 (so) both proteins are produced ; 5 use UV light (to detect fluorescence) ; 6 fluorescence shows, transcription / expression, of gene of interest (GOI) ; 7 AVP ; e.g. fluorescent proteins not harmful / have no effect on plant / do not affect quantity of desired protein produced	4
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Question	Answer	Marks
4(b)	any two from: 1 high / increase in, yield / productivity / quality or more food ; 2 increased income / reduce costs ; 3 can use herbicides / AW ; 4 less / no, competition ; 5 no need to plough / less labour intensive ;	2
4(c)(i)	complementary (base pairing between DNA strands) / described ;	1
4(c)(ii)	any two from: 1 use mRNA to make (ss)cDNA ; 2 fluorescent, tag / dye / substance / label, attached to cDNA ; 3 hybridisation / binding, of cDNA to probe ; 4 use, lasers / UV light, to produce fluorescence ; <i>plus</i> 5 idea of intensity of, fluorescence / light, indicates level of (gene) expression ;	3